```python
# Problem 1: Return all possible permutations of a string or array elements.
def permutations(text):

    if len(text) == 1:
        return [text]

    result = []
    for i in range(len(text)):
        current_char = text[i]

        remaining = ""
        for j in range(len(text)):
            if j != i:
                remaining += text[j]

        sub_permutations = permutations(remaining)

        for perm in sub_permutations:
            result.append(current_char + perm)

    return result

# Problem 2: Fill a 9x9 Sudoku board following Sudoku rules.
def is_valid(arr, row, col, num):
    for c in range(9):
        if arr[row][c] == num:
            return False

    for r in range(9):
        if arr[r][col] == num:
            return False

    startRow, startCol = 3 * (row // 3), 3 * (col // 3)
    for r in range(startRow, startRow + 3):
        for c in range(startCol, startCol + 3):
            if arr[r][c] == num:
                return False
    return True
def sudoku(arr):
    for row in range(9):
        for col in range(9):
            if arr[row][col] == 0:
                for num in range(1, 10):
                    if is_valid(arr, row, col, num):
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                        arr[row][col] = num
                        if sudoku(arr):
                            return True
                        arr[row][col] = 0
                return False
    return True

# Problem 3: Partition a string such that every substring is a palindrome. Return
all possible partitions.
    # i used some help from chatgpt becasue it was hard to understand
def is_palindrome(text):
    return text == text[::-1]
def palindrome_partitions(text):
    if not text:
        return [[]]

    result = []


    for i in range(1, len(text)+1):
        prefix = text[:i]
        if is_palindrome(prefix):
            for partition in palindrome_partitions(text[i:]):
                result.append([prefix] + partition)

    return result
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